Round 1 - SDE Intern - {DWEESHA CHAUDHARI}

Round 1 - Online assessment for SDE Intern role at FOG.

Please **copy this document**, replace your **name in the document title**, fill in the candidate details below, answer all questions, and **submit your completed document via the provided Google Form**.

Candidate Name:Dweesha Chaudhari

Form Link: https://forms.gle/J1G6V5LCFFsKyXpa6

Instructions -

- The assessment should be done solely by yourself.
- Submit the response to assessment within 3 days.

Tips to succeed -

- Pay attention to detail
- Think user first
- Be creative
- Use AI (Chat GPT, Midjourney, Claude, v0, etc…)

All the best and don't forget to have fun!

Question 1

Create a game selection interface based on the provided reference video. You can find the necessary card assets in the linked Drive directory. (Drive Link 🔗)

Interactive Features:

- When a game card is selected, play a placeholder (dummy) landscape video.
- Implement swipe functionality for navigating between cards.

User Experience:

- Ensure the UI is intuitive and user-friendly so that any player can easily navigate the game selection process.

Submission Requirements:

- Provide a link to the GitHub repository.

  ## https://github.com/dweeshaa/FOG_Assessment

- Include a live, hosted URL (e.g., Vercel, Netlify, GitHub Pages, or AWS).

  ## http://localhost:3000/question1/

- **Note:** Submissions without a live link will not be considered for review.

Question 2

Create a dynamic grid based on the reference video. The UI must include input fields for "Rows" and "Columns" (Minimum: 10x10). The pattern must scale dynamically to any grid size. (Drive Link 🔗)

- Provide a clear option to switch between **Pattern 1** and **Pattern 2**.
- If a player completes Pattern 1 successfully, they should automatically transition to Pattern 2.

Tile Mechanics & Visuals:

- Blue Tiles (Points): Players must collect these to gain points.
- Red Tiles (Danger): If touched, the tile must blink (White/Red), and the player loses 1 life.
- Green Tiles (Safe Zone): Neutral areas where the player can move safely.

Game Constraints & UI:

- Lives: The player starts with 5 lives.
- Timer: The player has 30 seconds to collect all blue points in each round.
- Win: All blue points collected within the time limit.
- Lose: Timer runs out or all 5 lives are lost.
- Display a clear "Win" or "Lose" message at the end of the session.

Submission Requirements:

- Provide a link to the GitHub repository.

  ## https://github.com/dweeshaa/FOG_Assessment

- Include a live, hosted URL (e.g., Vercel, Netlify, GitHub Pages, or AWS).

  ## http://localhost:3000/question2/

- **Note:** Submissions without a live link will not be considered for review.

Question 3

Have you built any game or worked on a game/gamification related project before? Tell us a little bit about what you did. Attach screenshots where required.

Answer

For this question, I am sharing a project completed as part of a **Digital Logic and System Design lab assignment**, where we built a basic car racing system using Verilog on a Basys 3 FPGA.

What I implemented:

- Developed a VGA-based display system (640×480 resolution) to render a car and road environment.
- Used ROM modules (BRAM) to store pixel data for both the background and car sprite.
- Implemented transparency logic to correctly overlay the car on the road by removing the sprite's background.
- Generated HSYNC and VSYNC signals using a VGA driver and verified correctness via simulation.

Core Features:

- **Car Movement:**
  The car can move left and right across the screen based on input.
- **Scrolling Background:**
  The road background was designed to move continuously, creating a simple animation effect and simulating forward motion.
- **Collision Detection:**
  Implemented basic collision detection logic between the car and road boundaries.
  - When a collision occurs, the system detects invalid positions.
  - A black screen is triggered on collision, acting as a failure/termination state.

Outcome:

- Successfully demonstrated a working hardware-level game prototype with:
  - Real-time rendering
  - Movement + animation
  - Collision handling
- Gained hands-on experience with digital design, memory management, and VGA-based graphics systems.

Scope:

This was a foundational implementation, focused on core mechanics (rendering, movement, collision), and can be extended further with scoring, obstacles, and gameplay logic.

Here is the screenshot of how the visual looked, when the car moved to left or right extreme and touched the green grass are then the program terminated and showed black screen.